

LLM-Assisted Entity Resolution: Summary of Findings and Current Status

2026-08-02

1 Purpose

This report describes how the entity-resolution pipeline works, explains each attribute-selection methodology it implements, and summarizes what the experiments so far establish. The methodology sections are complete; the results are condensed. Full per-run provenance and the detailed statistical analysis behind each finding are in the companion report dated 2026-08-01.

All numbers come from runs matching the pipeline’s current configuration: embedding model BAAI/bge-large-en-v1.5, matcher LLM google/gemini-2.5-flash via OpenRouter. Fodors–Zagat runs exclude the leaking `class` attribute (Section 4.4); runs whose LLM calls failed are excluded entirely (Section 5).

2 Pipeline Overview

The pipeline runs in five stages, each timed and recorded in a per-run JSON summary under `logs/<dataset>/runs/`.

Load and normalize. Both tables and the ground truth are read, ID columns are dropped, and every remaining field is passed through a normalization step that expands or standardizes address and directional abbreviations, and cleans up Unicode/HTML artifacts, so that superficial formatting differences do not by themselves create false mismatches later in the pipeline.

Embed and block. Each attribute column is encoded independently with the SentenceBERT model, the column vectors are concatenated and L2-normalized into one embedding per record, and multi-probe locality-sensitive hashing (random hyperplanes, with single-bit flips at query time) retrieves the top- k most similar Table-B candidates for every Table-A record. This reduces the pairwise comparison space from the full cross product down to a much smaller candidate-pair set before any LLM is involved.

Step 1 – global tuple condensation (optional). One attribute subset is chosen once for the whole dataset and applied to every candidate pair before re-blocking on the condensed attributes. This step is described in full in Section 3.1.

Step 1.5 – adaptive per-pair attribute selection (optional). Instead of one global subset, a small sample of candidate pairs is profiled by the LLM for per-attribute importance, and a hybrid predictor/retriever model transfers a *different* attribute subset to every candidate pair. This step is described in full in Section 3.2.

Match. One LLM call decides Yes/No per candidate pair, using only the attributes selected for that pair (all attributes if neither Step 1 nor Step 1.5 ran). Every response is cached on disk keyed by a hash of the model, prompt template, and record content, so repeated experiments over the same configuration do not re-spend tokens.

Metrics and logging. Precision, recall, and F1 are computed both at the candidate-pair level (only pairs that survived blocking) and end-to-end (relative to all ground-truth matches, so blocking misses count against recall), alongside token-usage accounting split by stage.

3 Methodology

3.1 Step 1: Global Tuple Condensation

The motivation for Step 1 is that showing an LLM every attribute for every pair wastes tokens on fields that are frequently empty, redundant, or simply irrelevant to the match decision, and can make the model’s decision worse by diluting the signal from the few attributes that actually distinguish entities. Step 1 picks *one* attribute subset for the whole dataset, used for every pair, via one of four strategies.

Manual selection. A fixed, user-supplied list of attribute names is kept and everything else is dropped, with no data-driven reasoning involved. It is the cheapest strategy to run (no LLM calls, no labeled data required) and is appropriate when a person can already tell from the schema which fields carry identity information, but it does not adapt if that judgment is wrong for some records.

Heuristic selection. Attributes are pruned using two label-free, corpus-level statistics: the fraction of missing values (an attribute that is empty in most records cannot be evidence of anything) and the fraction of unique values (an attribute that takes almost the same value everywhere carries little discriminative signal). Attributes that survive pruning are also *compressed*: long text fields are stripped of boilerplate words (via a stopwords/marketing-language regex list) and only high-signal tokens are kept – numbers, alphanumeric codes/identifiers, capitalized proper nouns, and unit-bearing measurements (e.g. "12 kg") – so that a lengthy description field still contributes its distinctive tokens without its full token count.

LLM-guided selection. A small labeled sample of matching and non-matching pairs is shown to the LLM one pair at a time, together with the question “which fields were most influential in deciding whether these two records match?”, and the model returns the smallest sufficient subset of attribute names for that pair. Responses are aggregated across the sample by *selection frequency* per attribute (what fraction of responses named that attribute), attributes are ranked by that frequency, and the final subset keeps every attribute above a frequency threshold (optionally capped to a top- k).

Supervised selection. For the same labeled sample, three field-level similarity features are computed per attribute for every pair – token-set Jaccard overlap, normalized edit-distance similarity, and an exact-match indicator – and a logistic regression classifier (features standardized, class-balanced) is trained to predict match/non-match from all attributes’ features at once. Each attribute’s importance is the maximum absolute regression weight across its three features, normalized so all attributes’ importance scores sum to one; the top-ranked attributes above a threshold (optionally capped to top- k) are kept. Because it learns from labeled evidence directly, this strategy tends to identify a very small, highly discriminative attribute subset (see Section 4).

Whichever strategy is used, the condensed table is then re-embedded and re-blocked (so blocking itself also benefits from the smaller, more focused attribute set) before matching.

3.2 Step 1.5: Pair-Adaptive Attribute Selection

Step 1.5 addresses a limitation of Step 1: even a well-chosen global attribute subset is not equally informative for every pair. For one pair an exact model number may already be decisive; for another, the deciding signal might be an author mismatch or a subtle title variation, while the model number field is missing or uninformative. Step 1.5 replaces one global subset with a *per-pair* subset, produced in two stages.

Profiling stage. A small sample of post-blocking candidate pairs (the “profile pairs”) is selected *without using ground-truth labels* by one of five samplers (described below). For each profile pair, the LLM is asked to score every attribute’s usefulness for the match/non-match decision on an integer 0–3 scale (0 = not useful/missing/generic, 3 = highly important or decisive), and the raw scores are normalized per pair into an importance vector that sums to one.

Transfer stage. A `HybridAttributeSelector` is fit on the profiled pairs’ label-free similarity features (token overlap, edit similarity, exact match, semantic similarity, numeric similarity, and an is-numeric indicator, computed per attribute) against their LLM-scored importance vectors. It combines two sub-models: a Ridge-regression *predictor* that maps a pair’s features directly to a predicted importance vector, and a cosine-similarity *retriever* that finds the k most similar profiled pairs and averages their importance vectors. The two signals are combined by a weighted average (0.5/0.5 in the current configuration). For every candidate pair (profiled or not), attributes are then selected in descending combined-importance order until either a fixed top- k is reached or cumulative importance crosses a threshold (0.8 in the current configuration, bounded between a minimum and maximum attribute count), with a fallback rule that force-includes at least one attribute tagged with a required “identity” role (e.g. title, name, model number) even if the learned ranking would otherwise drop it.

The five samplers below differ only in *which* candidate pairs get profiled by the LLM; the transfer stage is identical for all of them.

Random sampling. Profile pairs are drawn uniformly at random from all candidate pairs. This is the label-free baseline against which the other samplers are compared.

Similarity-stratified sampling. Candidate pairs are bucketed into low/medium/high terciles by their blocking-stage cosine similarity, and profile pairs are drawn from each bucket in fixed proportions (60% medium, 20% low, 20% high in the current configuration). This intentionally profiles some clearly similar pairs, some clearly dissimilar pairs, and a majority of ambiguous near-threshold pairs, on the reasoning that ambiguous pairs are where attribute importance is hardest to guess and most useful to learn from.

Diversity sampling. Profile pairs are chosen by farthest-first traversal in the label-free pair-feature space: starting from the pair farthest from the sample centroid, each subsequent pair is the one that maximizes its minimum distance to all previously chosen pairs. This favors broad coverage of distinct similarity “shapes” (e.g. pairs that agree on names but disagree on numbers, versus pairs that disagree on everything) over any particular similarity level.

Stratified diversity sampling. The same low/medium/high similarity strata as similarity-stratified sampling are used, but within each stratum, pairs are chosen by the same farthest-first diversity rule rather than uniformly at random. This is meant to combine both benefits: similarity-level balance and within-level feature diversity.

Importance-based sampling. Pairs are ranked by a label-free *importance proxy score* that estimates, without ground truth, how informative a pair is likely to be for learning attribute importance:

$$\text{proxy} = 0.50 \cdot \text{evidence_contrast} + 0.30 \cdot \text{similarity_uncertainty} + 0.20 \cdot \text{evidence_mixedness}$$

where `evidence_contrast` is the standard deviation of per-attribute agreement signals (favoring pairs where attributes disagree with each other, rather than all pointing the same way), `similarity_uncertainty` favors pairs whose blocking similarity is far from the sample’s typical value, and `evidence_mixedness` favors pairs whose average per-attribute agreement sits near 0.5 (neither clearly matching nor clearly non-matching). The top-scoring pairs (with random tie-breaking) become the profile sample.

4 Results and Findings

4.1 Attribute selection is worth doing

On the one dataset with a no-selection baseline, both mechanisms beat showing the matcher every field, and both spend fewer tokens doing it. This is the premise underlying the whole approach, and it holds.

Method	Attrs/pair	F1	Tokens	vs. no selection
No selection (<code>full</code>)	5.00	0.7849	802,477	—
Step 1.5, diversity (mean of 6 runs)	4.00	0.9222	775,828	+0.1373 F1, −3.3% tokens
Step 1, supervised	2.00	0.9537	658,185	+0.1688 F1, −18.0% tokens

Fodors–Zagat, leaking attribute excluded. Note that the simpler mechanism wins here: a single global two-attribute subset beats per-pair adaptive selection on both accuracy and cost. Section 4.2 qualifies what that comparison actually establishes.

4.2 The two mechanisms do not operate under the same supervision

The head-to-head above is not like-for-like, and the difference is large enough to change how the result should be read.

Step 1’s supervised strategy requires ground-truth labels. It calls `build_step1_labeled_pairs`, which draws its positive examples directly from the gold set — 50 positives and 50 negatives in the configuration used here — and trains a classifier on them. Step 1’s LLM-guided strategy needs the same labeled sample. Step 1.5, by contrast, consumes *no* labels at all: its profile pairs are chosen by label-free samplers and scored by the LLM, never by consulting ground truth.

Mechanism	Gold labels used	Fodors–Zagat	DBLP–ACM
Step 1, supervised / LLM-guided	50 pos. + 50 neg.	45.5% of gold	2.2% of gold
Step 1.5, all samplers	none	—	—

On Fodors–Zagat the asymmetry is severe: the ground truth contains 110 matches, and Step 1 supervised is trained on 50 of them. It is shown nearly half the answer key before it selects a single attribute.

Two consequences follow. First, Step 1 supervised is better understood as an approximate *upper bound* than as a deployable baseline — in a real deployment, the labels it consumes are exactly

what the pipeline exists to produce. Second, the labeled sample is not held out: those 50 gold pairs remain in the evaluation set, so a portion of the Step-1 score is measured on pairs whose attribute selection was fitted with knowledge of their labels. On Fodors–Zagat that overlap is 45.5% of the ground truth; on DBLP–ACM, where the gold set is much larger, it is 2.2% and correspondingly less serious.

This does not make the Step-1 numbers wrong, and it does not rescue Step 1.5, which still has to demonstrate a win on its own terms. But it reframes the comparison. The question Step 1.5 should be judged on is not whether it beats a method trained on the answer key — it is how close a fully label-free method gets to one. Judged that way, 0.9222 against 0.9537 on Fodors–Zagat with zero supervision is a materially different result from the same gap between two comparable methods.

The honest like-for-like baselines for Step 1.5 are Step 1’s *manual* and *heuristic* strategies, which are also label-free. Neither has current-pipeline results on any dataset, which is a real gap in the evaluation (Section 6).

4.3 The samplers differ in stability, not in accuracy

Running every sampler six times on Fodors–Zagat shows that most of the apparent ranking is noise. Among the top four, the differences in mean F1 are not statistically distinguishable. What *is* distinguishable is the spread: samplers that draw their profile pairs randomly vary an order of magnitude more than those that derive them deterministically.

Sampler	Mean F1	SD	TP range	Sampling
diversity	0.9222	0.0021	95–96	deterministic
stratified_diversity	0.9152	0.0042	95–97	deterministic
similarity_stratified	0.9110	0.0211	87–99	stochastic
random	0.8843	0.0429	79–96	stochastic
importance_based	0.8356	0.0063	78–81	deterministic

Two things follow. First, a single run of a stochastic sampler is not evidence: one **random** run can land anywhere from 79 to 96 true positives, spanning nearly the full gap between the best and worst methods. Second, the pipeline’s current default, **similarity_stratified**, is one of the unstable ones.

The one robust ranking result is negative: **importance_based** is reliably the worst sampler on this dataset, significantly so against all four alternatives. Since its sampling is deterministic, this is not one unlucky draw.

4.4 Fodors–Zagat contained a label leak

The **class** attribute is the ground-truth cluster identifier — it agrees on 110 of 110 gold pairs and is unique within each table. Step-1 supervised selection had been choosing it, which invalidated that result entirely. All Fodors–Zagat numbers here come from re-runs with the attribute excluded. DBLP–ACM was checked and is clean; Amazon–Walmart’s **original_id** was checked and does not leak. The leak was found by manual inspection; the pipeline has no automated guard.

4.5 The selection policy breaks on wide schemas

Step 1.5 keeps attributes until their cumulative importance reaches 0.8, capped at five. On Amazon–Walmart’s thirteen attributes this target is unreachable: using the LLM’s own importance scores,

the top five carry 0.612 of the mass on average and never more than 0.789, so reaching 0.8 would need 7.7 attributes. The cap therefore binds for 100% of pairs by construction, selection size is a constant 5.0, and the run tests nothing about adaptive selection. This is a property of normalizing importance over a wide schema, not a defect in the selector.

5 Two Defects Found in the Measurement Apparatus

Both have been fixed and covered by regression tests. Neither fix was in place for any result above.

5.1 Prompt field order was an uncontrolled variable

The selector emitted its chosen subset in predicted-importance order, and that order reached the prompt verbatim. Because predicted importance shifts with the profile sample, reseeding permuted the prompt even when it selected exactly the same fields. Conditioning every cross-seed comparison on DBLP-ACM:

Condition	Matcher changes its answer
Same attribute set, same order	0.10%
Same attribute set, different order	5.44%
Different attribute set	1.23%

Reordering the same fields changes the answer 54 times more often than re-sending an identical prompt — and more often than changing which fields are shown. Every Step-1.5 run has therefore been varying two things at once, and on this dataset the unintended one is the stronger. Both selection stages now sort into table-column order.

5.2 An unset token cap aborted runs while credit remained

The LLM client never set `max_tokens`, so every request defaulted to the model’s 65,535-token output ceiling. OpenRouter reserves credit against that declared ceiling, not against actual use, and returns HTTP 402 when the balance cannot cover it. The matcher in fact emits *exactly one* completion token per call (13,080 of 13,080 on DBLP-ACM; 2,665 of 2,665 on Fodors-Zagat). Requests were reserving roughly four orders of magnitude more budget than they used, and nine runs were destroyed this way. Explicit caps are now sent: 8 tokens for matching, 512 for profiling.

A related hazard remains partly open: a failed LLM call is scored as a non-match and the run still writes complete metrics, so a wholly failed run reports $F1 = 0.0000$ and a partly failed one reports a plausible-looking number. The sweep driver now discards these and halts; a directly launched run does not, and must be checked.

6 What Is Not Settled

The sampler ranking, on either dataset. The DBLP-ACM figures are single runs at one seed. Reseeding the one sampler that was successfully re-run moves F1 across 0.8847–0.9258, a spread of 0.0411 — wider than the 0.0370–0.0539 gap the single-seed table attributes to sampler choice. The headline result that `importance_based` wins on DBLP-ACM is therefore withdrawn pending repeated seeds: not contradicted, but unsupported. Replicate noise at fixed configuration is only 10 true positives, so the result may well survive; it has simply not been shown.

The attempted sweep completed five of thirteen runs before the budget ran out.

Cross-generation comparability. The canonical-order fix means no pre-fix run may be compared against a post-fix run. The Step-1.5 tables need regenerating in full rather than extending.

Whether Step 1.5 beats a label-free competitor. Every Step-1 result available comes from the supervised strategy, which consumes ground-truth labels (Section 4.2). The two label-free Step-1 strategies, *manual* and *heuristic*, have no current-pipeline results on any dataset, so the only comparison the evidence supports is against a supervised method — which Step 1.5 is not, and is not trying to be. Running those two baselines is cheap and would settle the central question of the thesis far more directly than another sampler sweep.

7 Status and Cost to Complete

Work is halted for a non-technical reason: the OpenRouter account is exhausted, at \$95.00 provisioned against \$95.25 consumed — \$0.25 overdrawn — and all calls return HTTP 402. Costs below use \$0.31 per million tokens, derived from this project’s own billing (48.1 M tokens over 46 runs in the past week against \$14.90 charged) rather than from list pricing.

Work item	Tokens	Est. cost	Runtime
Step-1 <i>manual</i> + <i>heuristic</i> , both datasets	9 M	\$2.79	1.1 h
Fodors–Zagat Step-1 baseline (re-run)	0.7 M	\$0.22	4 min
Fodors–Zagat sampler sweep, 5×5	20 M	\$6.19	1.7 h
DBLP–ACM <i>full</i> and Step-1 baselines	7.5 M	\$2.32	55 min
DBLP–ACM sampler sweep, 5×5	98 M	\$30.35	8.2 h
Total to complete the comparison	135 M	\$41.88	12 h
Amazon–Walmart, single run (blocked on policy)	60 M	\$18.58	4.3 h

Amazon–Walmart is blocked on the selection policy rather than on budget: until the cumulative-importance threshold is made scale-free, a run on its schema saturates and tests nothing. Spending on it first would buy another uninterpretable result.

8 Assessment

The pipeline works and attribute selection demonstrably helps. The uncomfortable findings are that the simpler mechanism currently posts the better number — a global two-attribute subset beats per-pair adaptive selection on Fodors–Zagat, on both accuracy and cost — and that the sampler comparison, which was the strongest apparent result, has shrunk under each round of scrutiny: first when a label leak was removed, then when repeated seeds showed the ranking was inside the noise, and finally when prompt field order proved to be a stronger driver than attribute choice.

That first finding needs qualifying rather than conceding. The two mechanisms are not solving the same problem: Step 1’s supervised strategy is trained on 50 gold matches, which on Fodors–Zagat is 45.5% of the entire answer key, while Step 1.5 uses no labels at all (Section 4.2). A method given half the ground truth should be expected to win. The result that matters is the size of the remaining gap — 0.9222 against 0.9537, with zero supervision — and that gap is small enough to be worth defending. Establishing this properly requires the two label-free Step-1 strategies, *manual* and *heuristic*, which have never been run on the current pipeline.

The rest of what survives is worth keeping. Attribute selection beats no selection by a wide margin. One sampler is reliably bad. And the samplers separate cleanly by stability rather than by

mean accuracy, which reframes what the contribution should be measured on — a more defensible claim than a ranking that reseeding erases.

The immediate need is API credit, approximately \$42, to regenerate the Step-1.5 comparisons under the corrected pipeline, plus the two label-free Step-1 baselines that would make the central comparison fair.